ECONOMICS · GRADE 12 · TERM 1 · WEEK 3

Why Government Intervenes: Market Failure, Macro Objectives, and the Policy Toolkit

The economic case for public administration in a market economy.

LOOKING AHEAD

Term 1 Outline — What We Will Cover

Unit — Public Finance and Policy (Weeks 3–6)

- Week 3 — Market failure, the government's policy toolkit (tax, subsidy, price controls, externalities) with full graphing practice, the four macro objectives — and SAU 1, the Unit 12.1A test on Money and Financial Institutions
- Week 4 — The state budget: its role, and its income and expenditure items
- Week 5 — Major types of taxes
- Week 6 — Fiscal, monetary and supply-side policy

Assessment

- Weeks 7–8 — reserved for the Unit Summative Assessment and feedback

LESSON 1 OF 3

Market Failure, and How Government Responds

LESSON 1

Learning Objectives

By the end of this lesson, you will be able to:

- Define market failure and explain why it justifies government intervention, using public goods, externalities, and information failure as three causes.
- Read a supply-and-demand graph to calculate the effect of a tax, a subsidy, a price ceiling, a price floor, and an externality — including deadweight loss, shortage, surplus, and the size of a tax or subsidy.

WARM-UP · 5 MIN · THINK–PAIR–SHARE

Who Pays for the Streetlight?

5 minutes

- A village wants to install streetlights along its main road. Everyone who walks that road at night benefits from the light, whether or not they paid toward it.
- Think for 1 minute: would a private company be able to profitably sell streetlights to individual households here? Why or why not?
- Compare your answer with a partner.

TEACHER INPUT · 8 MIN

Market Failure — Three Causes

- Market failure: when the free market allocates resources inefficiently, producing too much or too little of a good relative to the socially optimal level.
- Public goods — non-excludable (you can't stop someone using it) and non-rival (one person's use doesn't reduce another's), e.g. national defence, street lighting.
- Externalities — a cost or benefit that spills over onto a third party not involved in the transaction, e.g. pollution (negative) or a beekeeper's bees pollinating a neighbour's orchard (positive).
- Information failure — one party to a transaction lacks information the other has, leading to a poor decision, e.g. a used-car buyer unaware of a hidden fault.

GUIDED DISCOVERY · 10 MIN · GROUPS OF 3–4

Classify the Market Failure

10 minutes

- Your handout lists eight real-world scenarios. For each one, decide which type of market failure is present — a public good, a positive externality, a negative externality, or information failure — and justify your answer in one sentence.
- Some scenarios may show more than one type; be ready to explain your reasoning.
- Scenarios include: a private fireworks display visible to the whole neighbourhood; a factory releasing waste into a river used by farmers downstream; a used-car seller who knows about a hidden engine fault; national defence; a rural road that benefits an entire community; a patient trusting a doctor's treatment recommendation; and others on your handout.

QUICK CHECK · 5 MIN

Did You Classify Correctly?

Compare your answers with a neighbour on the two scenarios your teacher calls out, then be ready to justify your reasoning to the class.

1. Define market failure.

2. Give an example of a public good and explain why the market would under-provide it.

3. Distinguish a positive externality from a negative externality, using an example of each.

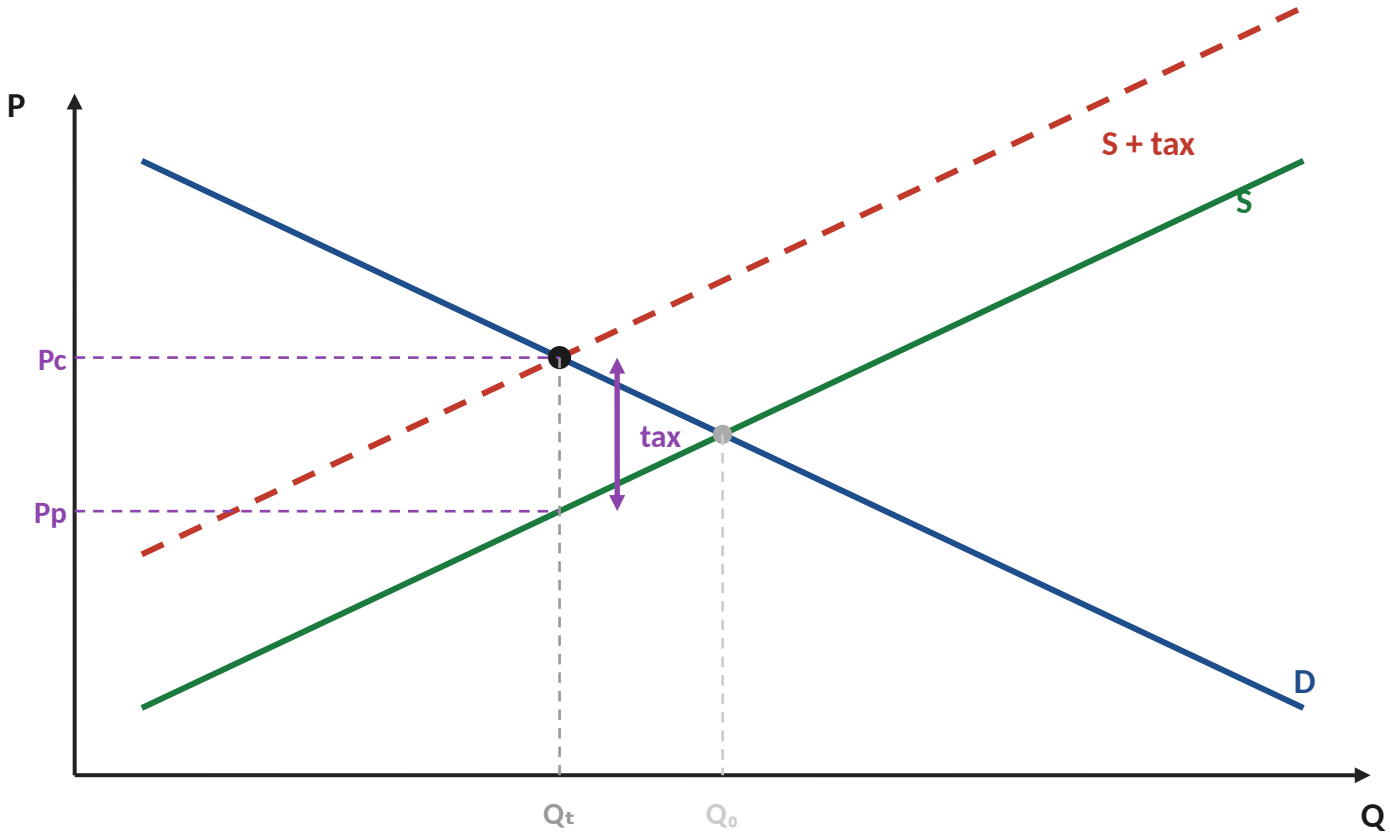
TEACHER INPUT · 5 MIN

From Diagnosis to Cure: The Government's Toolkit

- Indirect tax — raises the cost of producing a good with a negative externality, shifting supply left toward the socially optimal quantity.
- Subsidy — lowers the cost of producing a good with a positive externality, shifting supply right toward the socially optimal quantity.
- Price ceiling / floor — a legal maximum or minimum price, often used for equity reasons, but can create shortages or surpluses.
- Tradable permits — government caps total pollution and lets firms buy/sell permits, guaranteeing a pollution reduction at least cost.
- Direct provision — government supplies a public good itself (e.g. defence, street lighting) rather than relying on the market.
- Regulation — laws and standards that restrict harmful behaviour or require disclosure of information.

Indirect Tax

A per-unit tax shifts supply up (left) by the amount of the tax, raising the price consumers pay and lowering the price producers keep.



HOW TO CALCULATE

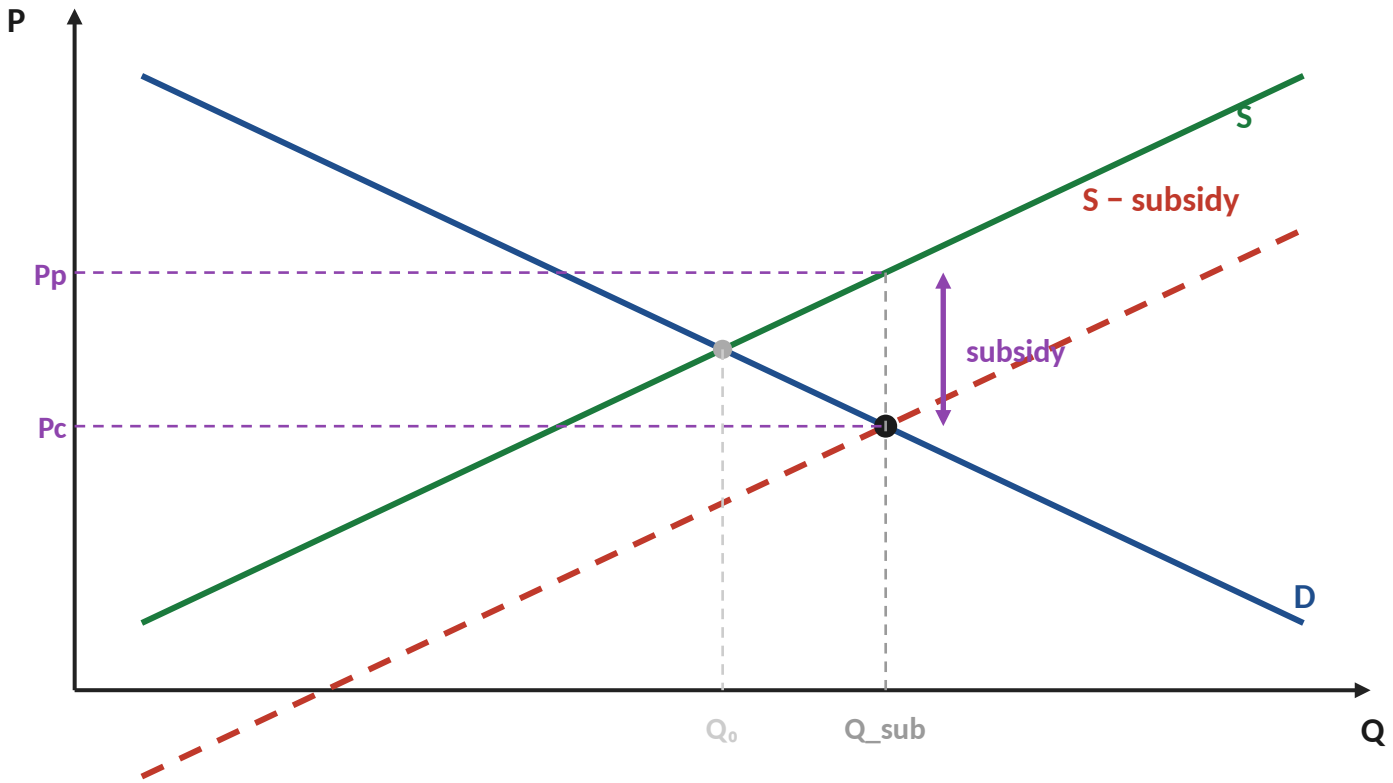
Tax per unit = Pc − Pp — the vertical wedge between the original supply curve and the tax-inclusive supply curve, measured at the new (lower) quantity Q_t.

Tax revenue = tax per unit × Q_t

Incidence Consumer burden = Pc − P_0 Producer burden = P_0 − Pp

Subsidy

A per-unit subsidy shifts supply down (right) by the amount of the subsidy, lowering the price consumers pay and raising the price producers receive.



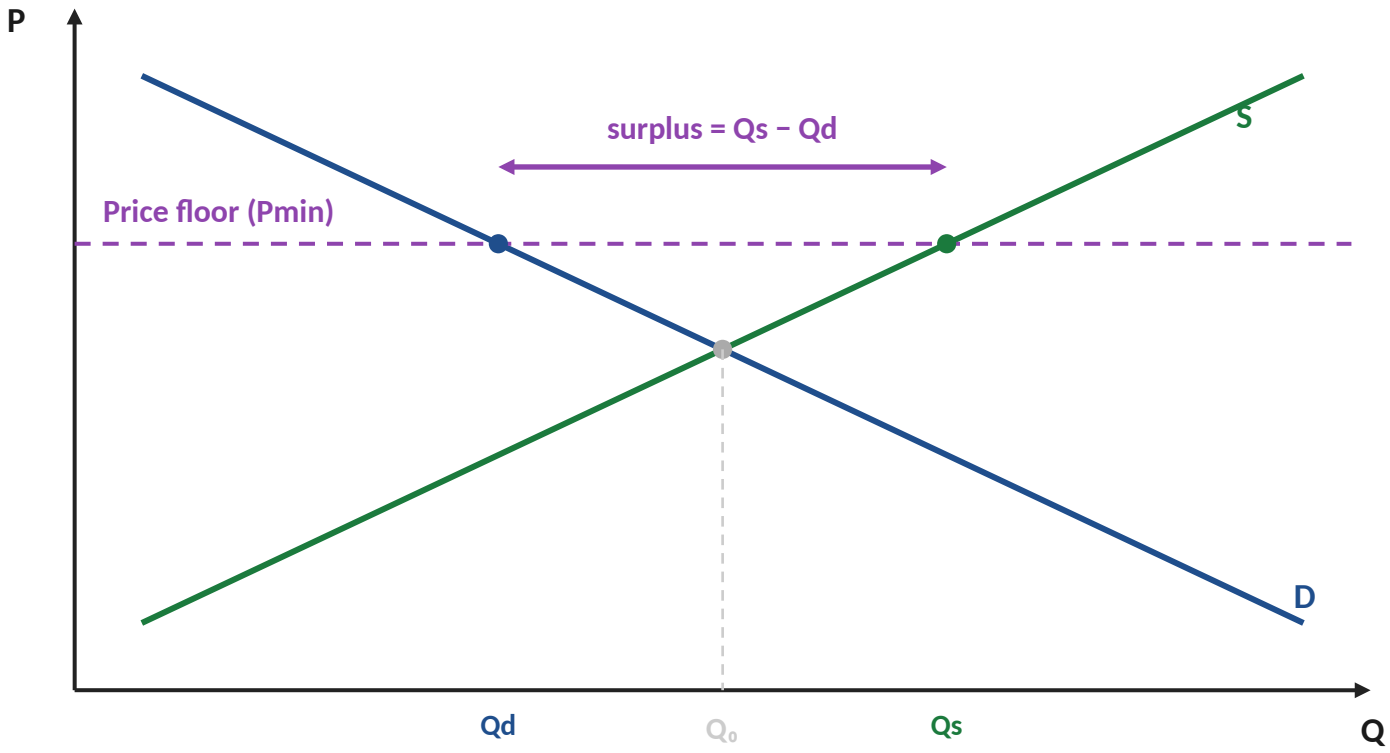
HOW TO CALCULATE

Subsidy per unit = Pp − Pc — the vertical wedge between the two supply curves, measured at the new (higher) quantity Q_sub.

Total cost to government = subsidy per unit × Q_sub

Price Floor

A minimum legal price set ABOVE the free-market price — used, for example, to guarantee farmers a minimum income. It creates a surplus.



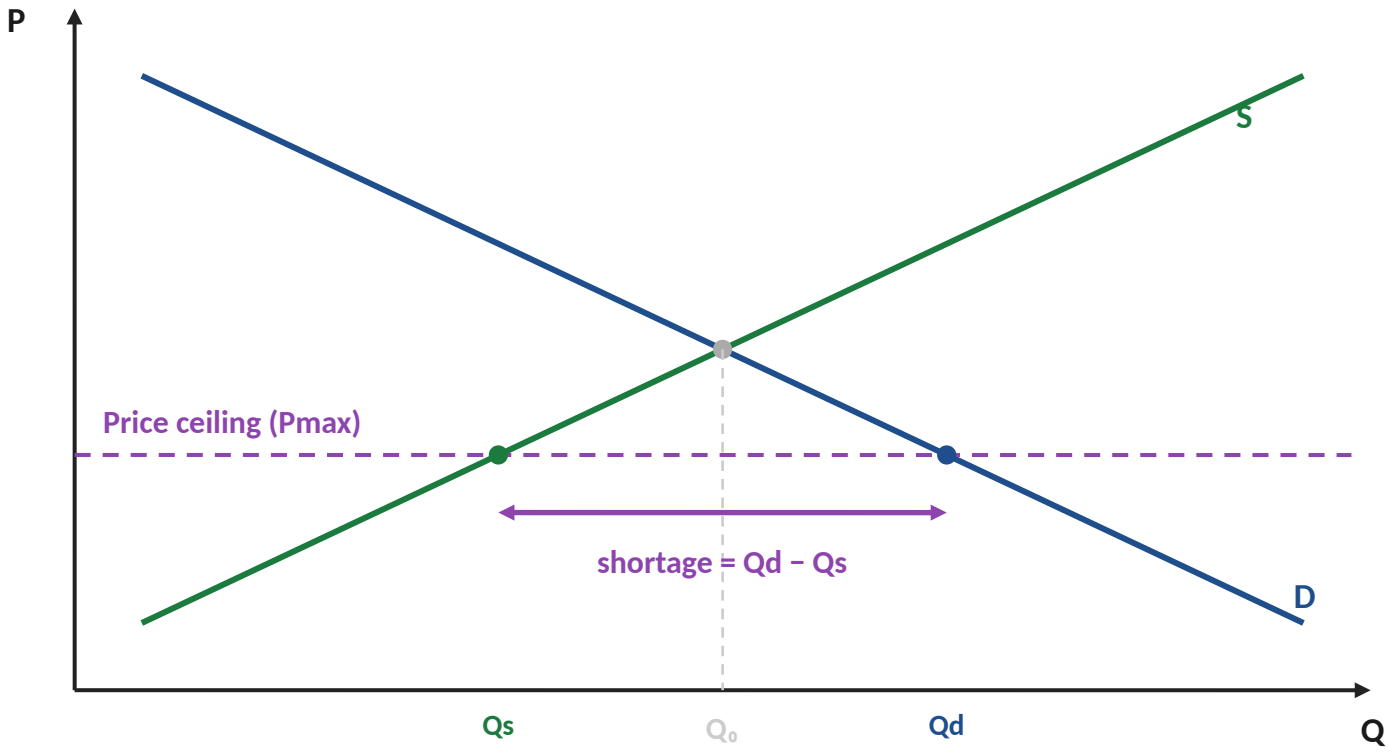
HOW TO CALCULATE

Surplus (excess supply) = Qs − Qd, both read straight off the graph at the floor price Pmin.

Note A floor only has an effect if it is set above the free-market equilibrium price.

Price Ceiling

A maximum legal price set BELOW the free-market price — used, for example, to keep a staple food affordable. It creates a shortage.



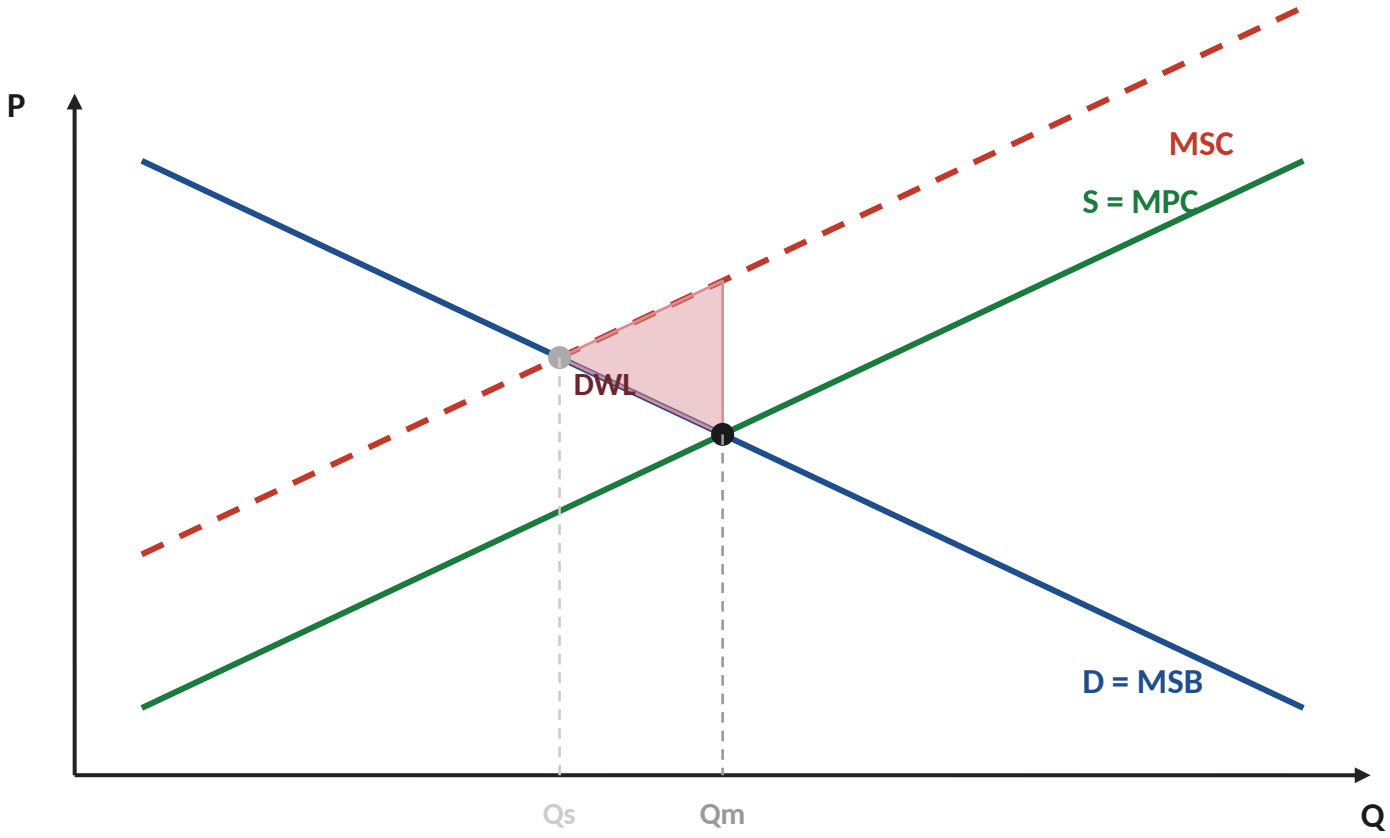
HOW TO CALCULATE

Shortage = Qd − Qs, both read straight off the graph at the ceiling price Pmax.

Note A ceiling only has an effect if it is set below the free-market equilibrium price.

Negative Externality

A production cost falls on a third party, so the marginal social cost (MSC) lies above the marginal private cost (MPC = supply). The free market overproduces.



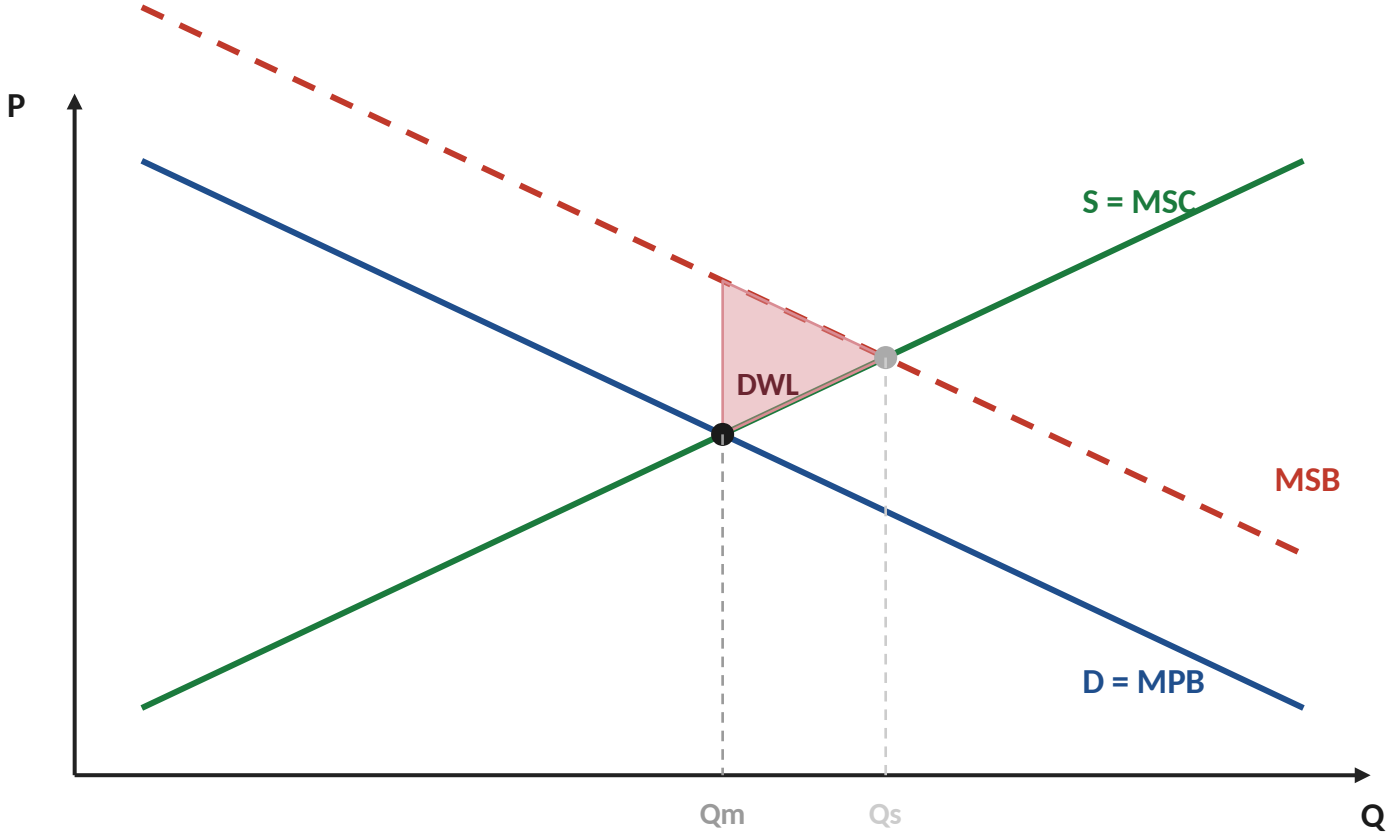
HOW TO CALCULATE

Deadweight loss = ½ × base × height = ½ × (Qm − Qs) × (MSC − MSB at Qm).

Note The triangle sits between the market quantity Qm and the smaller, socially optimal quantity Qs.

Positive Externality

A consumption benefit spills over to a third party, so the marginal social benefit (MSB) lies above the marginal private benefit (MPB = demand). The free market underproduces.



HOW TO CALCULATE

Deadweight loss = ½ × base × height = ½ × (Qs − Qm) × (MSB − MSC at Qm).

Note The loss here comes from under-consumption — too little is produced relative to the social optimum Qs.

TEACHER INPUT · 4 MIN

Government Failure — When Intervention Doesn't Help

- Government failure: when a well-intentioned intervention makes resource allocation less efficient, not more — intervention does not automatically improve on the market outcome.
- Imperfect information — government may not know the true external cost or benefit, so a tax or subsidy can be set at the wrong level.
- Unintended consequences — a price ceiling meant to help consumers can create shortages, queues, and black markets.
- Administrative and enforcement costs — designing, monitoring, and enforcing a policy uses resources that have their own opportunity cost.
- Regulatory capture and political self-interest — policy can be shaped by the industry it regulates, or by what is popular rather than what is efficient.

PLENARY · CONNECT IT BACK

From Market Failure to Policy Tool

1. Which tool would best address a factory polluting a river? Sketch the diagram you would use to justify your choice.

2. A price ceiling on bread is set below equilibrium. Explain, using your graph, why this creates a shortage rather than solving the underlying problem.

3. Give one example of government failure — a case where intervention could make the outcome worse rather than better.

LESSON 1 RECAP

Key Takeaways

- Market failure has three causes — public goods, externalities, and information failure — and is the economic justification for government intervention.
- A tax or subsidy shifts supply by the size of the per-unit amount; the price consumers pay and the price producers receive can be read directly off the resulting graph.
- A price ceiling below equilibrium creates a shortage; a price floor above equilibrium creates a surplus — both are measured at the controlled price.
- A negative externality causes overproduction and a negative-externality deadweight loss; a positive externality causes underproduction and its own deadweight loss — both are the triangle between the market and socially optimal quantities.
- Intervention is not automatically an improvement: government failure is a real risk of every tool in the toolkit.

BEFORE YOU GO

Exit Ticket

- Which of the six diagrams today did you find hardest to read? Why?
- One real-world example, from Kazakhstan or elsewhere, of a market failure and the tool you would use to correct it.

LESSON 2 OF 3

The Four Macro Objectives and the Case for Redistribution

LESSON 2

Learning Objectives

By the end of this lesson, you will be able to:

- State the four macroeconomic objectives of government policy and explain the goal of each.
- Explain the government's role in redistributing income, and the basic equity–efficiency trade-off.

WARM-UP · 5 MIN · PAIRS

If You Ran the Country for a Day

5 minutes

- If you were in charge of the country's economy for one day, which problem would you tackle first: rising prices, people out of work, slow growth, or something else?
- Compare your choice with a partner. Did you pick the same one? Why or why not?

TEACHER INPUT · 8 MIN

The Four Macroeconomic Objectives

- Economic growth — a sustainable increase in the economy's output over time, raising average living standards.
- Low unemployment — as many people who want to work as possible are in productive jobs.
- Low and stable inflation — prices rise slowly and predictably, so households and firms can plan.
- A sustainable balance of payments — the country's transactions with the rest of the world remain roughly in balance over time.
- Alongside these four, government also pursues income redistribution — using taxes and transfers to narrow the gap between rich and poor, which can come at some cost to economic efficiency.

JIGSAW — EXPERT GROUPS · 15 MIN

Become the Class Expert on One Objective

15 minutes · Home groups of 5

- Your group is assigned ONE objective: economic growth, low unemployment, low and stable inflation, balance of payments, or income redistribution.
- Read your scenario card. Discuss: what does this objective mean in practice, why does government care about it, and what happens to ordinary people if it is not achieved?
- Prepare a 60–90 second explanation, with your own example, ready to teach students who haven't seen your card.

JIGSAW — TEACHING ROUND · 9 MIN

Regroup and Teach Each Other

9 minutes · New mixed groups of 5

- Form new groups with one expert from each of the five original groups.
- Take turns teaching your objective — about 90 seconds each. Everyone else completes the notes grid on their handout as they listen.
- By the end, every group member should be able to explain all five objectives.

PLENARY · RAPID-FIRE CHECK

Five Objectives, Five Questions

1. Name the four macroeconomic objectives, plus income redistribution as a fifth policy concern.

2. Explain why low and stable inflation matters to an ordinary household.

3. Give one reason a government pursuing rapid economic growth might struggle to keep its balance of payments stable.

4. Give one way income redistribution can reduce economic efficiency.

LESSON 2 RECAP

Key Takeaways

- Government pursues four macroeconomic objectives: growth, low unemployment, low and stable inflation, and a sustainable balance of payments.
- Income redistribution is a further policy concern, balancing equity against economic efficiency.
- Achieving all of these objectives at once is difficult — progress on one can come at the cost of another.

BEFORE YOU GO

Exit Ticket

- Which of the five objectives/concerns do you think matters most in Kazakhstan right now? Why?
- One question you still have about how government tries to achieve these objectives.

LESSON 3 OF 3

SAU 1 — Summative Assessment for Unit 1

TODAY'S ASSESSMENT

SAU 1 — Money and Financial Institutions

- Covers Unit 12.1A: Money and Financial Institutions — the functions of money, financial institutions, and how interest rates are determined.
- 25 marks total, suggested time 25 minutes.
- Bring a pen. A calculator may be used for Question 6; write in pen and cross out errors instead of using correction fluid.
- No notes or textbooks. Work silently and independently for the full test period.

AFTER THE TEST

What's Next

- Once you finish, check your answers against the questions before handing your paper in.
- Marked papers and feedback will be returned before we move on.
- Next week: the state budget — its role, and its income and expenditure items.